

Self-Propelled Arm System (S.P.A.S.)

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Abstract—Autonomous waste collection has become an important application area for modern robotics and computer vision. This project presents the Self-Propelled Arm System (S.P.A.S.), an autonomous robotic platform capable of detecting recyclable objects, navigating toward them, grasping them using a robotic arm, and depositing them into a designated trash container. The system integrates a tracked mobile robot platform, a robotic manipulator, camera-based perception, and a YOLO-based object detection model with depth estimation for object localization. The project demonstrates how computer vision and robotics techniques can be combined to automate basic waste sorting tasks. Experimental evaluation shows successful object detection and pickup of bottles and cans while highlighting challenges related to object localization and robotic manipulation.

I. INTRODUCTION

Waste management is a repetitive task where autonomous robotic systems can provide assistance by reducing manual effort and improving efficiency. Recent advances in deep learning and computer vision have enabled robots to perceive and interact with their environment more effectively.

The goal of this project is to develop the **Self-Propelled Arm System (S.P.A.S.)**, an autonomous robot capable of detecting recyclable objects, approaching them, picking them up, and placing them into a trash container.

The project combines concepts from Cyber-Physical Systems and Robotics, including perception, autonomous control, object detection, and robotic manipulation.

The complete implementation is available in the project repository [1].

II. SYSTEM OVERVIEW

The developed system is built on the Hiwonder JetTank tracked robotic platform. The JetTank provides the mobile base and computational platform required for autonomous robotic applications.

The platform consists of a tracked chassis, onboard computing hardware, and expansion interfaces for additional robotic components. A robotic arm equipped with a claw was mounted on the platform to enable object manipulation. The

robot is controlled through a software pipeline combining computer vision, depth estimation, and motion control.

Unlike systems equipped with dedicated depth cameras, S.P.A.S. uses a standard RGB camera. Depth information is generated using a monocular depth estimation neural network, allowing the robot to estimate object distance from a single image.

The main hardware components are:

Component	Purpose
Hiwonder JetTank	Tracked mobile robotic platform
RGB Camera	Image acquisition for object detection
Jetson onboard computer	Running AI models and robot control software
Robotic arm with claw	Object grasping and transportation

The complete perception and control pipeline consists of four stages:

- 1) **Object detection:** RGB images captured by the camera are processed using YOLO to detect recyclable objects such as bottles and cans.
- 2) **Depth estimation:** After detecting an object, the corresponding image region is processed by a DepthNet model to estimate the object's distance from the camera.
- 3) **Navigation and positioning:** The estimated object position is used to control the JetTank movement and align the robotic arm.
- 4) **Manipulation:** The robotic arm performs the grasping and dropping sequence.

III. METHODOLOGY

A. Object Detection

Object detection is performed using the YOLO architecture (**You Only Look Once**). YOLO was selected because it

provides real-time object detection capabilities suitable for robotic applications [2].

A custom model was trained to identify recyclable objects, specifically cans and bottles. The detector receives camera images and outputs detected object classes together with their image coordinates.

B. Depth Estimation

Since the robot is equipped with a conventional RGB camera rather than a dedicated depth sensor, the system uses a monocular depth estimation approach.

After YOLO detects a recyclable object, the detected region of interest is passed to a DepthNet model. The model generates a depth map representing the estimated distance of pixels relative to the camera.

The estimated depth value of the detected object is then used to approximate its physical distance and assist the robotic arm positioning process.

This approach reduces hardware requirements but introduces additional uncertainty compared with dedicated depth sensors because depth is inferred from visual information.

C. Robotic Manipulation

The robotic arm performs a predefined sequence:

- 1) Move toward the detected object.
- 2) Position the claw above the object.
- 3) Close the claw.
- 4) Lift the object.
- 5) Move toward the trash container.
- 6) Release the object.

Robot manipulation techniques are based on common robotic modeling and control approaches described in [3].

IV. IMPLEMENTATION

The system software was implemented using Python and computer vision libraries.

The software architecture was separated into several modules:

- Camera input handling.
- YOLO inference.
- Depth estimation.
- Robot navigation.
- Arm control.
- System coordination.

This modular design simplified testing and allowed individual components to be improved independently.

The YOLO implementation was based on the Ultralytics framework [4].

A. Hardware and Software Integration

The JetTank platform provides the low-level hardware interfaces required for controlling the mobile base and receiving sensor information. The developed software extends the original platform capabilities by integrating custom object detection and autonomous pickup functionality.

The system combines ROS-based robot control with Python-based computer vision modules. The perception pipeline receives camera images, performs YOLO inference,

estimates object position, and sends movement commands to the robot controller.

The final implementation creates a complete cyber-physical loop:

Sensor Input → Perception → Decision Making → Actuation → Physical Interaction

V. RESULTS AND EVALUATION

Testing showed that the developed system successfully detected recyclable objects and performed autonomous pickup operations.

The robot was able to:

- Detect bottles and cans.
- Approach detected objects.
- Control the robotic arm.
- Successfully pick up objects.

However, several limitations were observed. Detection accuracy depended on environmental conditions such as lighting and object orientation. Additionally, precise robotic grasping remained challenging due to errors in object localization and mechanical positioning.

VI. CHALLENGES AND FUTURE WORK

The main challenges during development were related to reliable object detection and accurate robotic manipulation.

Training the YOLO model required suitable datasets and parameter tuning. Furthermore, connecting perception results with precise arm movement required calibration and repeated testing.

Future improvements include:

- Increasing the training dataset size.
- Using dedicated depth sensors.
- Implementing obstacle avoidance.
- Supporting additional waste categories.
- Improving grasp accuracy through feedback control.

VII. CONCLUSION

This project presented the development of the Self-Propelled Arm System (S.P.A.S.), a robotic platform for autonomous waste sorting.

By combining computer vision, object detection, depth estimation, and robotic manipulation, the system successfully demonstrated autonomous detection and pickup of recyclable objects.

The project provided practical experience in designing and implementing a Cyber-Physical System that integrates perception, decision making, and physical interaction.

REFERENCES

- [1] K. Kanev, "Self-Propelled Arm System (S.P.A.S.)." 2026.
- [2] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788. doi: 10.1109/CVPR.2016.91.
- [3] M. W. Spong, S. Hutchinson, and M. Vidyasagar, *Robot Modeling and Control*. Wiley, 2006.
- [4] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO." 2024.